

Prajwal Navada G P

prajwalnavada74@gmail.com | +91 8660228838 | Bengaluru, Karnataka | LinkedIn | GitHub | Portfolio

SUMMARY

Second-year CSE student (CGPA 9.2) with hands-on experience in software engineering, embedded systems, AI systems, and distributed applications. Developed full-stack platforms, real-time systems, and TinyML solutions on resource-constrained hardware. Strong foundation in C/C++, Linux, Operating Systems, Computer Networks, DBMS, and Data Structures & Algorithms. Interested in software engineering, systems programming, embedded software, AI systems, and Electronic Design Automation (EDA).

EDUCATION

Dr. H.N. National College of Engineering, Bengaluru 2024–2028
B.E. Computer Science & Engineering · Visvesvaraya Technological University | CGPA: 9.2

RELEVANT COURSEWORK

Data Structures & Algorithms, Operating Systems, Computer Networks, Database Management Systems, Object-Oriented Programming, Software Engineering, Distributed Systems

TECHNICAL SKILLS

Programming: C, C++, Java, Python, JavaScript

Systems & Tools: Linux (Ubuntu), Git, GitHub, MySQL, Firebase, Postman, Agile

Embedded & IoT: ESP32, Arduino (C++), MQTT, Sensor Fusion, Signal Processing, ThingsBoard, DHT11, MPU6050

AI & ML: TinyML (Edge Impulse), K-Means Anomaly Detection, Linear Regression, Predictive Analytics, LLM APIs

Software Engineering: React, Node.js, Express.js, Spring Boot, REST APIs, Socket.io, JWT, PWA

Core CS: DSA, OOP, DBMS, Computer Networks, Operating Systems, Real-Time Systems, Distributed Systems

PROJECTS

- EdgeAI-Predict — TinyML Predictive Maintenance System** GitHub
C++ · ESP32 · TinyML (Edge Impulse) · Signal Processing · MQTT · ThingsBoard
- Engineered a complete Edge AI system on a single ESP32 for real-time anomaly detection and predictive maintenance, eliminating dependency on gateway devices.
 - Designed a hybrid intelligent decision pipeline using moving-average filtering, linear regression forecasting, rule-based thresholds, and TinyML inference on-device.
 - Optimized C++ firmware for memory-constrained embedded execution, sensor fusion (DHT11 + MPU6050), and reliable TinyML inference on ESP32 hardware.
 - Implemented MQTT telemetry, real-time monitoring dashboards, anomaly alerts, trend visualization, and bidirectional RPC remote control.
- QuizLive — Distributed Real-Time Assessment Platform** GitHub | Live
React · Node.js · Socket.io · MySQL · JWT
- Built a real-time distributed system with synchronized quiz sessions, sub-100ms score propagation via Socket.io, dynamic leaderboards, and concurrent user support.
 - Designed a 10-table normalized schema; implemented JWT authentication, session validation, and anti-cheat mechanisms.
 - Developed a modular architecture focused on extensibility, maintainability, and performance.
- Auditorium Booking System — Multi-Role Enterprise Platform** GitHub
React · Vite · Node.js · Express · MySQL · Firebase FCM · PWA
- Architected a multi-role software platform with decoupled controller-service layers, role-based authorization, approval workflows, and conflict-detection logic.
 - Designed scalable relational database structures supporting Admin, Staff, and Principal workflows.
 - Integrated notifications, scheduled automation jobs, and PDF/Excel reporting pipelines.
- Smart Study — Automated Academic Resource Portal** GitHub | Live
React · Node.js · Google Drive API · Render
- Developed a deployed portal supporting 10+ subjects with automated catalog reconstruction from Drive metadata.
 - Integrated Google Drive APIs for content delivery, metadata synchronization, and administrative monitoring.
 - Implemented operational dashboards providing live visibility into platform resources and status.

CERTIFICATIONS

Programming in Java — NPTEL (IIT Kharagpur) · Elite + Gold

ACHIEVEMENTS & LEADERSHIP

- Led 3+ project teams, coordinating planning, implementation, testing, and delivery across software and embedded engineering initiatives.
- Active participant in technical and analytical quiz competitions, consistently ranking among top performers.

- Active contributor to technical communities and peer-learning initiatives; regularly participates in collaborative engineering sessions.